

Attributes of Precursor Cells

Char ¹	TSCs ²	Tr-TSC/PSCs ³	PSCs ⁴	EctoSCs ⁵	EndoSCs ⁶	MesoSCs ⁷	MSCs ⁸ (MesPCs)
Size	0.1-2 μ m	2-6 μ m	6-8 μ m	8-10 μ m	8-10 μ m	8-10 μ m	10-20 μ m
Trp blue ⁹	Pos ¹⁰	Pos	Neg ¹¹	Neg	Neg	Neg	Neg
Viab T-sts ¹²	+4°C	+4°C	+4°C	+4°C	+4°C	+4°C	+4°C
Viab PM-sts ¹³	30+ days	30+ days	14 days	3 days	3 days	3 days	1 day
Cryo Ag ¹⁴	DMSO ¹⁵	DMSO	DMSO	DMSO	DMSO	DMSO	DMSO
Con Ag ¹⁶	7.5% v/v	7.5% v/v	7.5% v/v	7.5% v/v	7.5% v/v	7.5% v/v	10%
# Cryo ¹⁷	1-10 B ¹⁸	1-10 B	1-10 M ¹⁹	1-10 M	1-10 M	1-10 M	1-10 M
Op Fr T ²⁰	-80°C	-80°C	-80°C	-70°C	-70°C	-70°C	Liq N ₂ ²¹ / -196°C
Fr Pro ²²	Slow	Slow	Slow	Slow	Slow	Slow	Flash
Viab T-lts ²³	-80°C	-80°C	-80°C	-70°C	-70°C	-70°C	Liq N ₂ / -196°C
Thaw P ²⁴	Fast	Fast	Fast	Fast	Fast	Fast	Fast
Thaw T ²⁵	37°C	37°C	37°C	37°C	37°C	37°C	37°C
Recovery	>98%	>98%	>98%	>98%	>98%	>98%	>95%
Viab PM-lts ²⁶	20+ yrs ²⁷	20+ yrs	20+ yrs	22+ yrs	22+ yrs	22+ yrs	22+ yrs
P-Sol Org ²⁸	Yes	Yes	Yes	Yes	Yes	Yes	Yes
P-Skel Musc ²⁹	Yes	Yes	Yes	Yes	Yes	Yes	Yes
P-Adipose Tiss ³⁰	Yes	Yes	Yes	Yes	Yes	Yes	Yes
P-CTs ³¹	Yes	Yes	Yes	Yes	Yes	Yes	Yes
P-B Mar ³²	Yes	Yes	Yes	Yes	Yes	Yes	Yes
P-Blood ³³	Yes	Yes	Yes	Yes	Yes	Yes	No
Species ³⁴	M,Rt,Rb,F,Cn O,Cp,P,B,E,H	M,Rt,Rb,F,Cn O,Cp,P,B,E,H	M,Rt,Rb,F,Cn O,Cp,P,B,E,H	M,Rt,H	M,Rt,H	M,Rt,Rb,F,Cn O,Cp,P,B,E,H	Rt,H
Con Hib ³⁵	No	No	No	Yes	Yes	Yes	Yes
Growth	Sus ³⁶ /Adh ³⁷	Adherent	Adherent	Adherent	Adherent	Adherent	Adherent
Substrate	Coll-I ³⁸	Coll-I	Coll-I	Coll-I	Coll-I	Coll-I	Coll-I
SFM ³⁹	Quiescent	Quiescent	Quiescent	Quiescent	Quiescent	Quiescent	Quiescent
No GFac ⁴⁰	Quiescent	Quiescent	Quiescent	Quiescent	Quiescent	Quiescent	Quiescent
Inhib Fac ⁴¹	Respond	Respond	Respond	Respond	Respond	Respond	Respond
Prolif Fac ⁴²	Prolif ⁴³	Prolif	Prolif	Prolif	Prolif	Prolif	Prolif
Prolif Rt ⁴⁴	12-14 hr	12-14 hr	12-14 hr	18-24 hr	18-24 hr	18-24 hr	Ds – Ws ⁴⁵
Pop Dbl ⁴⁶	>300	>300	>400	>400	>400	>690	50-70
ProgreFac ⁴⁷	No	No	No	No	No	No	Yes
Induc Fac ⁴⁸	Yes	Yes	Yes	Yes – Lin only ⁴⁹	Yes – Lin only	Yes – Lin only	Yes – Lin only
Lineages	3 + Sp ⁵⁰	3	3	1	1	1	NA
Commit ⁵¹	Tr-TSCs	PSCs	EctoSCs, EndoSCs, MesoSCs	Ectodermal Lineage Cells	Endodermal Lineage Cells	Mesodermal Lineage Cells	Fat, Cartilage, Bone
# Cs ID ⁵²	68	67	63	13	11	39	3
Karyo ⁵³	Normal	Normal	Normal	NYD	Normal	Normal	NYD
Genes Expressed	Telom ⁵⁴ , Bcl-2, Nanog, Nanos, CXCR4	NYD	Telom, Oct-4 ⁵⁵ , Sonic hh ⁵⁶	NYD	Telom	NYD	
Cell Surface Markers ⁵⁷	CD66e, CEA, CEA-CAM-1	CD66e, CEA, CEA-CAM-1, CD10, SSEA	CD10, SSEA	CD90, Thy-1, CD56, MHC-I	CD90, Thy-1, MHC-I	CD90, Thy-1, CD13, MHC-I	CD105, CD117, CD166, MHC-I
Animal Models	Repair: Skeletal Muscle, Cartilage, Bone, Vessels, Inhib Scar Tissue, Parkinson disease,	Repair: Skeletal Muscle, Cartilage, Bone, Vessels, Inhib Scar Tissue, Parkinson disease,	Repair: Skeletal Muscle, Cartilage, Bone, Vessels, Inhib Scar Tissue, Parkinson disease,	Repair: Parkinson disease,	Repair: Pancreatic Islets	Repair: Skeletal Muscle, Cartilage, Bone, Vessels, Inhib Scar Tissue	NYD

	Myocard Infarction, Pulmonary Disease, Pancreatic Islets	Myocard Infarction, Pulmonary Disease, Pancreatic Islets	Myocard Infarction, Pulmonary Disease, Pancreatic Islets				
Human Models	PD ⁵⁸ , AD ⁵⁹ , MS ⁶⁰ , ALS ⁶¹ , CIDP ⁶² , CP ⁶³ , TBI ⁶⁴ , Strk ⁶⁵ , Sciatica, MD ⁶⁶ , LI ⁶⁷ , SLE ⁶⁸ , CVD ⁶⁹ , BMT ⁷⁰ , FR ⁷¹ , IPF ⁷² , COPD ⁷³ , Asth ⁷⁴ , TD ⁷⁵ , Diab ⁷⁶ , LF ⁷⁷ , HL ⁷⁸ , W ⁷⁹ , Ag ⁸⁰ , OI ⁸¹ , Mskl ⁸²	PD, AD, MS, ALS, CIDP, CP, TBI, Strk, Sciatica, MD, LI, SLE, CVD, FR, IPF, COPD, Asth, TD, Diab, LF, HL, W, Ag, OI, Mskl,	PD, AD, MS, ALS, CIDP, CP, TBI, Strk, Sciatica, MD, LI, SLE, CVD, FR, IPF, COPD, Asth, TD, Diab, LF, HL, W, Ag, OI, Mskl,	PD, AD, MS, ALS, CIDP, CP, TBI, Strk, Sciatica, MD,	NYD	MD, LI, SLE, CVD, FR, W, Ag, OI, Mskl,	NYD

Legend: Attributes of Precursor Cells.

- 1, Char = characteristics tested
- 2, TSCs = totipotent stem cells
- 3, Tr-TSCs = transitional totipotent stem cells
- 4, PSCs = pluripotent stem cells
- 5, EctoSCs = ectodermal stem cells
- 6, EndoSCs, endodermal stem cells
- 7, MesoSCs = mesodermal germ layer lineage stem cells
- 8, MSCs = (Caplan's Mesenchymal Stem Cells = tripotent progenitor cell that only forms fat, cartilage, and bone [Pittenger et al., 1999])
- 9, Trp blue = Trypan blue staining of cells to denote absence (positive) or presence (negative) of machinery to pump out dye within cytoplasm of cells
- 10, Pos = positive
- 11, Neg = negative
- 12, Viab T-sts= viability temperature for short term storage in which >95% of population is viable
- 13, Viab PM-sts = viability post mortem in short term storage
- 14, Cryo Ag, cryogenic agent tested
- 15, 99.9% ultra-pure DMSO, dimethyl sulfoxide
- 16, Con Ag, concentration of cryogenic agent
- 17, # Cryo, number of cells cryopreserved for optimum cellular recovery
- 18, B, billion
- 19, M, million
- 20, Op Fr T, optimum freezing temperature
- 21, Liq N₂, liquid nitrogen
- 22, Fr Pro, freezing process, rate at which the temperature was reduced during freezing of the cells
- 23, Viab T-lts = viability temperature for long term storage in which >95% of population is viable
- 24, Thaw P, thawing procedure
- 25, Thaw T, thawing temperature, this was accomplished using a 37°C water bath. We used the change in medium color as an indicator of whether the cells were frozen (yellow) or thawed (salmon).
- 26, Viab PM-lts = viability post mortem in long term storage (last time cryopreserved cells were tested for viability)
- 27, Yrs, years
- 28, P-Sol Org, presence in solid organs: trachea, lungs, esophagus, stomach, liver, intestines, spleen, brain, pancreas, kidney, urinary bladder, meninges, testis, tongue, and thyroid.
- 29, P-Skel Musc, presence in skeletal muscle
- 30, P-Adipose Tiss, presence in adipose tissue
- 31, P-CT, presence in connective tissues, i.e., skeletal muscle connective tissues (epimysium, perimysium, endomysium), dermis, heart (epicardium, myocardium, endocardium), granulation tissues, and associated connective tissues of periosteum (bone), perichondrium (cartilage), nerve sheaths (epineurium, perineurium, endoneurium), adipose tissue, ligaments, tendons, blood vessels.
- 32, P-B Mar, presence in bone marrow
- 33, P-Blood, presence in blood
- 34, Species, mammalian species in which precursor cells were identified: M, mouse; Rt, rat; Rb, rabbit; F, feline; Cn, canine; O, ovine; Cp, caprine; P, porcine; B, bovine; E, equine; H, human
- 35, Con Hib, contact inhibited at confluence forming a single layer of cells
- 36, Sus, suspension cultures
- 37, Adh, adherent to a substratum
- 38, Coll-I, preferred substratum is type-I collagen
- 39, SFM, serum-free defined media
- 40, No GFac, no growth factors added to medium
- 41, Inhib Fac, inhibitory factors added to medium, i.e., leukemia inhibitory factor (LIF) and/or anti-differentiation factor (ADF)

42, Prolif Fac, proliferation factor added to medium, i.e., platelet-derived growth factor-BB (PDGF-BB) to stimulate cellular proliferation

43, Proliferation of tested stem cells, assayed by amount of DNA per well as assessed by the DNA portion of the ELICA procedure

44, Prolif Rt, proliferation rate of the cells. The cells were assessed during log phase growth

45, Ds – Ws, days to weeks

46, Pop Dbl, population doublings

47, ProgreFac, progression factor (2 ng/ml insulin) added to the culture medium to accelerate phenotypic expression of lineage-committed progenitor cells

48, Induc F, induction factor added to the medium to assay for the ability to form multiple phenotypes in culture. We routinely used dexamethasone at 10^{-6} , 10^{-7} , 10^{-8} , 10^{-9} , 10^{-10} M as a non-specific induction agent, that dependent on concentration, would multiple cell types within the same well. We also used specific induction agents such as: Sk-MMP (skeletal muscle morphogenetic protein) for skeletal muscle, Sm-MMP for smooth muscle, Car-MMP for myocardium, Adip-MMP for adipocytes (fat cells), C-MMP (chondrocytes), BMP-2 (endochondral ossification), VEGF (endothelial cells), EPO (erythrocytes), HGF (hepatocytes), KGF (keratinocytes), BDGF (neurons), and conditioned medium from other differentiated cell types, i.e., sperm, to attempt to elicit a response.

49, Yes – Lin only, cells would respond positively to inductive factor but only with respect to their differentiation lineage. In other words, mesodermal inductive factors would only induce differentiation in cells of the mesodermal developmental lineage. Similarly, ectodermal inductive factors would only induce differentiation in cells of the ectodermal developmental lineage and endodermal inductive factors would only induce differentiation in cells of the endodermal developmental lineage. The tripotent progenitor, MSCs can only be induced by factors inducing fat, cartilage, and bone.

50, Sp, sperm

51, Commit, particular commitment into a specific developmental lineage.

52, # Cs ID, number of cell types identified using the ELICA procedure with antibodies and histochemical stains (including enzymes as negative staining controls). The number of cell types identified was not dependent on the differentiation capabilities of the cells, but rather the limited number of cell specific assays that were present in the laboratory at the time of analysis.

53, Karyo, karyotypic analysis

54, Telom, presence of the telomerase enzyme

55, Oct-4, expression of Oct-3/4 gene

56, Sonic hh, expression of the sonic hedgehog gene

57, Cell Surface Markers, cluster of differentiation markers and cell surface markers that were positively expressed on the surface of the cells as assessed by FACS analysis and antibody staining during the ELICA procedures. The CD markers used for the analysis included CD1a, CD2, CD3, CD4, CD5, CD7, CD8, CD9, CD10, CD11b, CD11c, CD13, CD14, CD15, CD16, CD18, CD19, CD20, CD22, CD23, CD24, CD25, CD31, CD33, CD34, CD36, CD38, CD41, CD42b, CD45, CD49d, CD55, CD56, CD57, CD59, CD61, CD62e, CD65, CD66e, CD68, CD69, CD71, CD79, CD83, CD90, CD95, CD105, CD117, CD123, CD135, CD166, Glycophorin-A, MHC-I, HLA-DR-II, FMC-7, Annexin-V and Lin

58, PD, Parkinson disease

59, AD, Alzheimer's disease

60, MS, Multiple Sclerosis

61, ALS, Amyotrophic Lateral Sclerosis

62, CIDP, Chronic Inflammatory Demyelinating Polyneuropathy

63, CP, Cerebral Palsy

64, TBI, Traumatic Brain Injury

65, Strk, Stroke

66, MD, Muscular Dystrophy

67, LI, Limb Ischemia

68, SLE, Systemic Lupus Erythematosus

69, CVD, Cardiovascular Disease

70, BMT, Bone Marrow Transplant

71, FR, Facial Reconstruction

72, IPF, Interstitial Pulmonary Fibrosis

73, COPD, Chronic obstructive pulmonary disease with forced expiratory volume in one second (FEV1) both above 50% (controlled pharmacologically) and below 50% (controlled with supplemental oxygen)

74, Asth, Asthma

75, TD, Thyroid (Hashimoto's) Disease

76, Diab, Diabetes Type-I

77, LF, Liver Fibrosis

78, HL, Hair Loss

79, W, Wrinkles

80, Ag, Aging

81, OI, Orthopedic Injuries

82, Muscle Injuries